Aman Verma

SOFTWARE DEVELOPMENT ENGINEER BACKEND INTERN

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WORK EXPERIENCE

Software Engineer Intern Jan 2026 - Jun 2026

LexoLegal *Remote*

- Architected async **REST APIs** in **FastAPI**, **SQLAlchemy**, and **PostgreSQL** (**SQL**) behind an **Nginx** proxy on **Linux**, implementing **JWT** authentication containerized via **Docker** with **Git** and **GitHub Actions** CI/CD.
- Deployed 'Munshi', an autonomous assistant in **Pydantic AI** and **FastAPI**, resolving multi-turn inquiries with **sub-2s latency** via structured tool-calling loops.
- Prevented race conditions from rapid concurrent requests by designing **Celery** workers with atomic **Redis** distributed locks and non-blocking 10s countdown requeues.
- Automated judicial data ingestion across 25 High Courts and 743 District Courts via **Celery** and **Bash** on **Linux**, utilizing **Selectolax** extractors for batch **PostgreSQL** updates.

Freelance Developer Oct 2024 - Jan 2025

Webxela *Remote*

- Implemented Square POS **OAuth 2.0** authentication and webhook ingestion in **Spring Boot** and **Spring Data JPA**, automating merchant onboarding in under 2 minutes with idempotent transaction verification.
- Developed a concurrent trademark search engine in **Kotlin**, **Spring Boot**, and **PostgreSQL**, leveraging **Coroutines** with custom thread-pool dispatchers, rate-limited connections, and indexed trigram queries.

PROJECTS

ncert-rag-evals | *Python, Jupyter Notebooks, SQLite FTS5, ChromaDB, Pytest* GitHub

- Evaluated an empirical retrieval benchmark across 17 STEM textbooks in **Python** and **Jupyter Notebooks**, implementing custom hybrid search in **SQLite FTS5** and **ChromaDB** to profile 8 strategies on CPU latency.
- Proved that fusing BM25 with dense embeddings via Reciprocal Rank Fusion (RRF) achieved **86.9% Recall@5 in 8.9ms on CPU**, matching LLM query expansion while cutting latency by **99.8%**.
- Built a document parser indexing 4,429 section-aware chunks into **SQLite FTS5** and **ChromaDB**; validated statistical significance via paired McNemar tests (**p = 0.0021**) across 92 unit tests in **Pytest**.

doubtless | *Pydantic AI, FastAPI, Celery, Redis, SQLite, OpenTelemetry, Logfire* GitHub

- Engineered an interactive lecture doubt-solver in **Pydantic AI** and **FastAPI**, fusing blackboard video frames with 45s dialogue windows to stream real-time solutions for visual queries via SSE.
- Designed an asynchronous 4-stage media pipeline (PipelineRunner) in **Celery** and **Redis** that transcodes HLS streams, slices 16kHz audio (<15MB/hr), and transcribes in <10s via Groq Whisper-Large-V3.
- Constructed the async data layer with thread-safe **SQLite** WAL repositories and **Redis** caching; instrumented distributed tracing via **OpenTelemetry** and **Logfire** across 151 unit tests.

vectorless-rag | *Python, Jupyter Notebooks, Matplotlib, Pydantic AI, Ollama, ChromaDB, PyMuPDF* GitHub

- Analyzed dense vector search against LLM page indexing on benchmark documents using **Pydantic AI**, configuring **Ollama** as a local testbed alongside cloud LLMs across 18 benchmark queries.
- Benchmarked retrieval trade-offs in **Jupyter Notebooks** using **Matplotlib**: plotted the **2.7x token cost** and **2x latency** tradeoff (**100% retrieval hit rate** and **0.85 citation precision**).
- Created a zero-framework document parser in **Python** using **PyMuPDF** and **ChromaDB**, streaming page-aligned windows with bounded memory (**<200MB**) cutting extraction latency by **40%**.

TECHNICAL SKILLS

Languages & Core: Python, Kotlin, SQL, Bash, Linux

Backend & Systems: FastAPI, Spring Boot, Celery, Redis, PostgreSQL, SQLite, SQLAlchemy

Infrastructure & APIs: Docker, Nginx, REST APIs, OAuth 2.0, JWT, Git, GitHub Actions

Testing, Data & Observability: Pytest, OpenTelemetry, Logfire, Pydantic AI, ChromaDB, Jupyter Notebooks

LEADERSHIP & ACTIVITIES

- Delivered 3 technical sessions on 'Generative AI Unlocked' and 'Level Up with AI', training 150+ college students on foundational LLM concepts and modern AI developer tools.
- Organized Smart India Hackathon internal rounds across 2 editions; coordinated problem statement curation, mentoring, and judging logistics for 20+ teams and 120+ participants.

EDUCATION

Veer Bahadur Singh Purvanchal University 2023 - 2027

Bachelor of Technology in Computer Science & Engineering *Jaunpur, India*